INFORMATION SERVICES, A KEY CONCEPT IN CORPORATE PORTALS

Ingela Bruce, Kerstin Forsberg

AstraZeneca, Softronic, Viktoria Institute
ingela.bruce@softronic.se, kerstin.l.forsberg@astrazeneca.se

Abstract

The paper use the concept of information services as a key concept to better understand the often misinterpreted concepts of intranet and corporate portal. The argumentation is based on three case studies of corporate portals and a literature review from a reflective practitioners perspective. Three different types of information services are being identified and discussed: 1) corporate communication owned, 2) business function driven, 3) transparency enabling. The paper also discusses key competences needed to develop the information services.

Keywords: Intranet, portals, information services

1. INTRODUCTION

The purpose of this paper is to shed some light to the difference between traditional intranets and corporate portals. The two concepts are in current practice used interchangeable and sometimes used on false pretences to argue for a re-brand of a traditional intranet to become a "portal".

In this paper we combine three different cases of portal implementations with reflections based on our own practice in the IT and Corporate Communication departments respectively during the last decade in Intranet development. As reflective practitioners (Schöön, 1996) we experience a gap of understanding the differences and similarities between intranets and portals in many corporations of today.

The paper is organised in the following way: In the next two sections, we give a background and an overview of related research and introduce associated concepts such as web services. Thereafter we describe the three different cases. In section four we discuss the three different types of information services identified in the cases. We also discuss the future professional skills needed for providing strategic and business value adding services delivered through corporate portals. In the concluding discussion, in section six , we summarise the paper.

2. BACKGROUND

The early adopters of intranets were a mixture of IT and Corporate Communication personnel who often built these first intranets from their own web pioneer idealism. The knowledge and skills needed when developing intranets at this time were basic and did not resemble a classical IT project. The usage of the technology has extensively increased and the term Intranet is now familiar to most corporations. Intranet is though sometimes still used and perceived as a non-strategic application: Intranets are "merely pumping information between different organisational units" (Hendriks, 2001) with a vast range of different presentational pages that strengthen the already existing departments or sites as organisational silos. This "ironic outcome" as described by Newell et al (1999) has led to the search for new possibilities in developing intranets to a more strategic and business adding value concept. This enhancement work is often being performed by personnel in various parts of the organisation. That is IT personnel and by people in communication, as well as in marketing, product development and production.

In the late 90ies the term 'portal' arose and became a way to offer Internet users a web-based single point of access to on-line information with a combination of search engine functionality and categorised navigation. When the term portal was taken inside the organisational firewalls, the term portal got companions: "corporate portal", "corporate information portal", "knowledge portal", "business portal" and "enterprise information portal". According to Detlor (2000) a portal's primary function is not merely to contain information as traditional intranets but to provide access to information and functionality already available elsewhere in the organisation.

In the market "blurr" of concept there is also a new type of application called portal frameworks which is a software application that allows many different visual components to be presented in a homogeneous "box:ified" style. These visual components are often given different names by the vendors; gadgets, iViews, adapters, components. In this paper we use the word *portlets* to describe these components. They present *information services,* which is the concept we explore in this paper.

The development of existing intranets are often just face-lifts to re-brand parts of, or whole, traditional intranets to become "portals". In this paper we discuss the consequences of the use of information services when large corporations want to move the content on their traditional intranets to become part of "portals".

3. RELATED RESEARCH AND KEY CONCEPTS

The possibility of intranets to provide support for Knowledge Management (KM) with knowledge as the key factor have been discussed by Stenmark (2002). Further Newell et al (1999) describe how the intranet reinforced the existing functional and national barriers with electronic knowledge silos. These two papers are examples of a matured discussion in the community of researchers of the role of and consequences of intranets, something we have in vain searched for regarding the concept of portals and information services.

The main aim of the corporate portals has so far been described as to "promote the gathering, sharing, and dissemination of information throughout the enterprise." (Detlor, 2000). Corporate portals have been characterized through different generations of portals (Eckerson, 1999) and named different due to different types of functionality (Dias, 2001). Eckerson (1999) distinguishes between four different generations of portals: Referential (content management, mass dissemination of corporate information and decision support), Personalized (customized distributed content), Interactive (collaborative features of the portal) and Specialized (connect corporate applications with the portal). The term "enterprise information portal" (EIP) was coined in November 1998 (Shilakes and Tylman, 1998).

> "*Enterprise information portals are applications that enable companies to unlock internally and externally stored information and provide users a single gateway to personalized information needed to make informed business decisions."* (Shilakes and Tylman, 1998).

While Internet portals are virtually dead, many corporations have now invested in portal framework software with the ambition to establish a new corporate information infrastructure (Nielsen, 2003). Vendors of portal framework software promise a broad range of capabilities such as consistent user interface, single sign-on security for applications, presentation layer for future applications and access from any network-connected device.

Dahlbom (2002) argues for the transition from information systems to information services. The technology "will provide occasional services to be bought and consumed in a much more flexible way" (Dahlbom, 2002). Applying this description on the discussion on intranet vs. portal, the intranets mirror the information systems in supporting corporations. In contrast the portals enable information services to be consumed by each individual. For example, we do not develop a information system for buying books, we use a information service offered by Amazon.com. A service that provide us with a straightforward process of identifying, buying and paying a specific book or CD. If our decisions involves more uncertainties and a higher degree of complexity, the e-commerce service present a number of adaptive features such as suggestions of books and CD:s based on purchasing and navigation patterns, our own as well as others. Mathiassen and Sørensen (2002) propose a task based theory for information services in terms of four different types: computational services, adaptive services, networking services and collaborative services. These four types combines Mintzberg's four different types of decision making depending on the degree of uncertainty and complexity with Gutek's theory of services in general divided into encounter and relationship.

Framing the information service concept from a technical perspective highlights several other concepts such as web services, portlets, personalisation, customization and role-based filtering.

The concept of web services has emerged as a buzzword in the information technology industry, but is to be recognized as the next natural step in the evolution of application integration. From a

technical perspective web services is a combination of XML based protocols[1] to develop, describe, package and discover functionality. From a business perspective web services makes it possible to market, provide and consume services.

Portlets are embedded services delivered as visual components in a portal. The portlets rely on the portal framework software to function, e.g. access to user profiles, access to the information object that represents the context in which the portlet appears as a box. Portlets are often branded by different vendors portal framework software using different names such as components, gadgets, iViews, adaptors etc.

The functionality provided by these portlets can be different, ranging from basic "screen-scraping" services[2] to full-fledged packages of business transactions. A trend is to combine web services with the idea of portlets. A proposed standard in this space is the "Web Services for Remote Portals" (WSRP[3]).

Personalisation, one of the core concepts in the portal world, could be described as the ability to intelligently improve the user experience (CXO Media, 2001). This personalisation of the portal provides each individual user with adequate information services adopted to serve the tasks for each individual user. This is not to be confused with the concept of customisation that in a portal context refers to the ability for the user to change the appearance of portlets, define placement of selected portlets and also to select from a predefined set-up of portlets.

The role-based filtering of content where the user gets what he/she needs is the push possibility from the organisational perspective. Depending on what sort of role you have in the company you get the informational access that the organisation, project or process chart shows.

4. CASES

4.1 Method

Using the Schöön (1996) perspective as reflective practitioners, we have framed the reality of our practical experiences gained from several intranets and portals projects through the years. This frame awareness has shown us certain dilemmas in the concepts of intranet and corporate portal that we here try to challenge and discuss based on three specific cases.

We have conducted structured, open-ended interviews with key stakeholders of portal initiatives in three corporations with focus on its content and how it differs or resembles traditional intranets respectively corporate portals. The stakeholders represent a mix of IT and Communicational background. These three cases represent different phases of portals: from early concept phase to

---

[1] Examples of XML protocols included in the web service concept: UDDI (Universal Description, Discovery and Integration) for discovery, WSDL (Web Services Description Language) for description, SOAP (Simple Object Access Protocol) for packaging of web services. See 'Web Services Acronyms, Demystified', XML.com, January 09, 2002

[2] Screen-scraping services means capturing a picture of the screen with the data it wants, and trying to figure out how to interpret the picture back into data and extract relevant data to be able to present them in an new context.

[3] Web Services for Remote Portals (WSRP) define a standard for interactive, user-facing web services that plug and play with portals. Providers of web services using the WSRP standard determine how their content and applications are visualized for end-users and to which degree customisation of the appearance may be allowed.

matured set of information services. The cases are described as they existed when the research took place in early spring 2002.

Each interview was conducted face to face and lasted approximately one hour. The authors have previous knowledge of the cases, either as consultant or as employee. Once transcribed, the interviews were analysed. The interviews were supported by a list of the following questions:

- Could you in short describe your portal (background, vision, origin of initiative, main content, planned content)
- How was the implementation planned?
- How did the implementation occur? (technical, content, cultivation, changed attitudes)
- Did you use parallel processes; if yes; how did they correspond to each other?
- What made your portal a success?
- What could have been done differently?

4.2 Holmen

Holmen is a solid, century old international company within pulp and paper with headquarters in Stockholm, Sweden. Holmen has a strong position in the market and shall grow faster than the market.

The decision of the Holmen portal originated from the former CEO who in 2000 created a vision of the modern way of knowledge sharing and creation. The CEO assigned the CIO for the mission of building the portal. A pre study started October 2000. In February 2001 a decision was taken to buy licenses for portal software in order to build the platform. In April 2001a new project leader was assigned and the project focus changed to cover also enhancement of the work of understanding the concept within the organisation. The vision of the portal was still the same as in the beginning - to be the natural working tool for every employee within Holmen. The place for your information, services and business-systems that you need in your daily work with only one log in.

The content of the portal was divided into two main parts. The real portal's name is "Holmen Views Portal" with single sign on and personalisation has approximately 300 users and contains shared services as knowledge sharing through different parts of the organisation. The second part. beside the portal, is "Holmen Views Open". This is the intranet entrance open for everyone without special log in. The users can here access all the different existing intranets within the organisation. No one of these have yet been melted down to a partly reappearance as functions/services.

The implementation of the Holmen portal concept is a process still in action. Its main obstacles have been the difficulties in the understanding of the portal concept. as well as the acceptance by the by the employees. But thanks to the open Intranet access – "Holmen Views Open" – the obstacles to gain acceptance of the real portal concept have been a bit easier.

The success of the implementation of the portal – "Holmen Views Portal" – can be attributed to the different communities or fora. Within Holmen today there exists today ca 30 different fora. Each forum has its own coach that is responsible for the vision, content and further expansion of the forum. One promising planned new content that has a possibility of success is the Forum for new employees.

Things that could have been done differently are definitely the start of the long-term building of acceptance in the organisation. The willingness of changes of attitudes towards using the new tool is what takes most time and is the hardest one to have easy and quick wins on. As the project leader puts it: the technology is 20% the rest is changes of attitudes and acceptance of the portal concept.

4.3 SAS

SAS – Scandinavian Airline Services key task is to offer air travel with base in its home market in Northern Europe. The SAS Groups goal is to be a company characterised by employees who: care, can be trusted, are progressive and professional.

The background of the portal within SAS starts with the vision by the CIO who in 1999 envisioned that SAS personnel should work smarter, which among other things meant access to information for SAS personnel from anywhere. The vision was named NAiS – New Ways of Working within SAS - with three main sponsors: Corporate Communication, Human Resources and Corporate IT. In the NAiS project it was decided that the existing intranet – FREiA should be shut down but could for a restricted amount of time be accessible through the new portal. The main content of the SAS portal was hence already by start the existing intranets. Four news channels for the different regions within SAS were gathered and made accessible in the portal with the Corporate Communication department as owner. A real and unique success from the core business perspective has been the "My Function" part – "Crew Services" - for the flying personnel. Here every member of the crew can by single sign on and personalization obtain his/her tailor-made information and tailored accessed applications that is in use before and after each flight.

The preparation of the implementation started as a pre study during autumn 1999 when many preparation workshops, seminars and road shows took place. These activities all aimed to strengthen the awareness, understanding and acceptance of the portal that was to be implemented. An example of such a project was measurement of attitudes and use of IT tools that were performed with three different user groups with different levels of knowledge of IT tools. This project's result was dual: the different groups had in the end the same high knowledge and the acceptance of the portal concept was equally high since the groups willingly wanted to be focus groups of the portal.

The implementation from a technical point of view part took place Jan 1 2000. The content was gathered in a library part called "All Services". Here the users self tailored what sort of content they wanted to be shown in their own "My Services" part.

Other important things that strengthen the use of the portal was the seriously world wide historical events that occurred during autumn 2001 that had impact on the airline business. Hence the employees in SAS instantly saw the impact of the portal as a news medium and the portal was now the main communication tool within SAS. As said earlier, another main reason for the success of the portal was the implementation of the "My Function" part, "Crew Services". It was a well-planned designed project with e.g.. special enhancements of knowledge through "floor workers" at the different airports.

But the first success of the portal in 2000 was the fast flow of news as well as the function "People Search". The People Search was an application that merged information from three different database sources, among them the SAP database (HR System). This application was used immensely and the first week after technical implementation the personnel responsible for the SAP HR database for the "People Search" application got hundreds of mails from employees finding themselves in the wrong organisation, their name and or telephone number wrong. Here, the transparency got the normal learning curve to change in a pace that was unfamiliar to the administrators of the content owner. More resources soon solved this problem in order to get the content correct.

Despite the success some things could have been done differently. The main publishing tool has not yet been presented and the lack of this have postponed some of the acceptance at least by all

employees that have to use old web publishing tools. A total understanding of the portal in the whole organisation has yet to come – at all levels including the senior management team.

4.4 AstraZeneca

AstraZeneca is one of the largest pharmaceutical companies in the world with an R&D organisation of 10.000 employees world in five different countries. The research process involves a large number of different scientific disciplines and research communities. The main field of research in AstraZeneca is focused within seven therapeutic area such as oncology, cardiovascular and gastrointestinal, respiratory and infection.

An initiative called the Clinical Informatics Forum has recognized a number of interesting capabilities in portals from an end-user perspective to increase creativity and effectiveness by enabling clinical researchers to share scientific information. A key concept in the project is the idea of supporting communities of researchers. The Apolipoprotein Research Project been selected as a first case for such a community.

The vision for the initiative is to enhance creativity, facilitate decision making and increase efficiency by enabling clinical researchers to exploit scientific information globally and to enlarge personal networks.

One key issue for the Informatics Forum portal is the navigation among clinical documents and clinical data across processes and therapeutic areas while making interaction between researchers easy. It exploits tools to share, search, navigate and reuse information from a research area's perspective that corresponds to an exploring research issue, rather than the needs of an operational activity. The scientific information resides in a wide range of data sources and is made available through different applications that are designed to serve a specific function, most often supporting operational activities. Information flow and integration among these applications are today limited.

The first case in the Informatics Forum project is Apolipoprotein research domain which examines the risk markers for death and myocardial infarctions related to abnormal lipids. The project initiative is still in its early days, exploring both capabilities offered by portal frameworks, online access to documents using the concept matching technology of Autonomy[4] and online access and analyze of clinical data from different studies. In the Apolipoprotein research domain a technology called Topic Maps[5] has been evaluated to describe knowledge structures in a machine-processable way and associating them with information resources.

The planned implementation strategy was not yet defined when the interviews was conducted.

---

[4] Autonomy software uses advanced pattern matching techniques utilising Bayesian Inference and Shannon's Information Theory. It identifies the patterns that naturally occur in a text, through usage or frequency of words. For more information see www.autonomy.com

[5] Topic Maps are an ISO/IEC standard that enables the structuring and navigating of large amounts of information. The Topic Map technology is based on the ways of structuring knowledge that can be found in indexes, thesauri, glossaries and semantic networks.

5. DISCUSSION

The three case studies show divergences in the content of the different portals and in the implementations strategies. In all three cases, a recurring theme is the changing perception of the concepts of intranets and portals. In the discussion we try to shed light on this blend of understanding of what differs a intranet from a portal viewed from the perspective of how the content is being produced and used.

We have chosen to discuss the information services offered through portals in three different cases by dividing them into three different types: 1) corporate communication owned, 2) business function driven, 3) transparency enabling. The authors base this discussion also from practical intranet/portal experience in other large Swedish based organisations.

The services that are business function driven and acts on behalf of the user are the ones that in the best way show the differences between a portal/intranet and a traditional intranet. We choose to use the separating factors in the three cases in order to present ideas for future areas to study. In the last part we also discuss key competences needed to develop these different types of information services.

5.1 Corporate communication owned information services

The use of the news channels provided by the portal in SAS case during the serious world wide historical events that occurred during autumn 2001 is an example of the widespread acceptance of the portal by all employees. This is an example of a traditional intranet service, now reachable through a personalized and role-based portal.

If wanting to clear the differences between intranet and portals this type of news channels content is as stated above more a similarity than a difference. However, the vendors of portal frameworks often set up showcases including such news feed channels for e.g. news, weather, stock-exchange rate, to demonstrate the value of portals. As practitioners, and advocates of portals as the next step in the evolution of intranet, we often find this type of sales pitch a problem since this kind of functionality does not distinguish portals from matured intranets based on web content management solutions. We have found it valuable to consider this type of corporate owned information services as a presumption when selecting a portal framework. However, a key feature is the possibility for different parts, and different levels, in the organisation to add their content to be delivered through this type of news service.

Furthermore, the type of personalisation that makes this type of information services a success has to offer much more than a way of adjusting the look and feel of a portal view. A recurring theme in both the SAS and Holmen case are that the use of visual modules within a portal user interface restricts the design of the user interface and many users experiences this as a “dull and boxy” design. This user experience is important when getting a broad acceptance and also in the planning the process for implementation of a portal framework. In some cases, employees working with intranet can experience the portal boxy design as an insult compared to their previous possibilities with creative design of their own intranet pages. Our experience, from different implementations of portal frameworks and developing information services, this has to be met in a dialogue showing the everyday business values of portal delivered information services for employees in different situations, helping them out with specific tasks.

5.2 Business function driven information services

The success of the My Function part of the portal initiative at SAS, Crew Services, for the flying personnel is an example of an package of functionality that acts on behalf of the user. Making pieces of functionality available like this makes individuals more efficient and better informed. All the needed functionality and information of the logistics related to every flight and to every

crewmember already existed in different information systems and manuals. To be able to provide all this in a situation and task-based context, the specific value-adding function bites and information pieces had to be made available as easy to consume information services.

Dahlbom (2002) describes information services as "the idea of being given the information needed at the time when you need it. It is the idea of having a servant who does all the hard work of collecting information, having overview, etc, so that you don't have to worry about that. All you have to do is ask for and receive the services." (Dahlbom, 2002)

We have found these business function driven information services the key factor for success in many portal initiatives. However, these information services have to be identified, uncovered and developed, re-packaged, made accessible and branded. The result is explicit business values for each individual. This work demands for a organisational endurance that in every case we so far have seen is lacking.

These types of information services packaged as portlets is more seldom included in the demonstrations prepared by vendors of portal frameworks since they are much more complex, highly depending of the type of industry as well as unique procedures, information structures and information systems in each organisation.

Furthermore, this type of functional driven information services calls for a combination of different professional skills. To be able to identify and specify the prerequisites and outcome of the service, and to be able to formulate the business value, a combination of core business knowledge, business analysis and evaluating impact is needed. While, the technical aspect of uncovering the specific functionality and information from existing information systems and data storages, and to develop interfaces and new parts requires knowledge about application development and databases. The important activities of getting buy-in, re-packing, branding and documenting the information service and training end users, demands for communication and education skills.

5.3 Transparency enabling information services

The examples of the 30 different fora in the example from Holmen, and the research domain of Apolipoprotein represent examples of trying to make different locations, organisational silos and information systems, transparent for people sharing the same focus or situation. Often this is typical for processes, projects and communities crossing existing organisational hierarchies, geographically distances and disparate skills.

We can identify two different types of information services enabling such a transparency, i.e. 1) Sharing information resources in a way that makes them exploitable. This is addressing the problem of shortage of adequate and understandable information while being overloaded with unfocused data and documents. 2) Networking and communicating to be able to create new knowledge together within the community of individuals people sharing the same focus or situation.

We have so far in our work with different portal initiatives been exposed to the challenges of the first type information services, i.e. sharing information. The need for having a taxonomy as part of the portal infrastructure to be able to structure, categorise, aggregate and navigate information has recently been highlighted and the words taxonomy and ontologies have become buzzwords. In short, ontology describes all concepts that exist within a world/domain. Taxonomy describes the relationships between the concepts. They both could be described as "a shared and common understanding of a domain that can be communicated between people and heterogeneous and widely spread application systems" (Fensel, 2001). To develop and maintain taxonomies and ontologies we need a combination of domain expertise and experiences of information science.

The second kind of information services enabling networking and communicating is something we often hear as the objective for many portal initiatives and this is a type of services that many research domains have been exploring (Brown and Duguid 1991, Wenger 1998). However, from our experience few implementations address these kind of functionality so far. Instead they use the word community to describe the possibility of sharing of information within a listed group that people can join. The word community, which has different distinct meaning among researchers, distinguishing between community of practice (CoP), community of interest or a community of knowledge, has be become high fashion in many organisations and is used for all types of groups e.g. projects, departments or teams.

5.4 Key competences

The transition from information systems to information services (Dahlbom 2002) demands for a rethinking also when applied to the development of services for a portal. The individual user of a service is mostly not responsible for the development of the service. Instead the user is the consumer of the service and gets his/her work done more efficient when using the service. A consultative approach is demanded with professionals trained to identify needs and desires, calculate on business values and return of investment, develop them technically and coordinate the development. Also, parallel non-technical issues that adds business value are equally important for the success. Examples of this are for buy in in-house marketing, user information and or training and the continuous work of identifying new compatible services.

The consultative approach when identifying and building an information service. A clear understanding of the business needs is basic as well as a user friendly and orientated development approach. The result is that someone else than yourself creates a service that you benefit from. The other parts have identified your need and in return they benefit from of your usage of the service.

However, the initial selection and implementation of a portal framework software is much more like a traditional IT project. It has to deal with issues such as performance, network load, application integration, robustness, standard compliance and purchase routines. This phase is characterised by tasks related to implementation of information systems, project control and structure. While the phase of utilising the portal framework to deliver portlets aggregated in one large portal or through a federated model of portals, is a open ended process of stimulating cultivation (Ciborra, 2000) of information services in which the usage of the portal tends do drift.

6. SUMMARY

Without taking full advantage of the possibility of using information services as the key in reaching the right target group or individual with adequate content and functionality, a launch of a new intranet or portal will be merely a face lift of a traditional intranet. The services that are business function driven and acts on behalf of the user are the ones that in the best way show the differences between a traditional intranet and a corporate portal. The transition demands for a consultative perspective within the organisation in order to recognize, develop and make new information services available.

The intra-organisational battles and their interactions and its outcome when planning, building and deploying an intranet or a portal would be interesting to study in more depth. The interaction is often a mixture of in-house political strings and often takes place in or between core business versus corporate services such as IT, Communication and Human Resources. We mean that the implementation, deployment and cultivation of portal projects is a mixture of traditional IT respectively communicational skills as well as deep business process knowledge.

In the future, we would like to study portal implementation strategies, organisational aspects and their effectiveness in different organisations. A potential framework for catagorisation of portlets delivered through the studied portals could be the task based framework for information services proposed by Mathiassen and Sørensen (2002).

6. REFERENCES

Brown J.S. and Duguid, P. (1991), Organizational Learning and Communities-of-Practice: Toward a Unified View of Working, Learning and Innovation. Organization Science, Vol. 2, No. 1.

CXO Media, June 2001, 'Understanding Portals: a Business Person's Guide to Enterprise Portal Terms & Business Impacts'

Dahlbom, B (2002), 'From systems to services", working paper, available at www.informatics.gu.se/~dahlbom

Detlor, B. (2000), 'The corporate portal as information infrastructure: Towards a framework for portal design', Int'l J. of Info. Mgmt., vol.20, pp.91-101.

Dias, C.(2001), 'Corporate Portals: A litterature review of a new concept in information management', Int'l J. of Info. Mgmt., vol.21, pp.269-287.

Eckerson, W. (1999), 'Plumtree blossoms: New version fulfils enterprise portal requirements', Patricia Seybold Group, Boston, M.A.

Fensel, D., (2001), 'Ontologies: Silver Bullet for Knowledge Management and Electronic Commerce', Springer-Verlag, Berlin

Heffner, R. and Ramos, L (2001), 'Key Portal Strategy Issues: Definition, Infrastructure and Payback', Giga Information Group

Hendriks, P. (2001) "Many Rivers to Cross: From ICT to Knowledge Management Systems", Journal of Information Technology,16 pp. 57-72.

Mack R., Y. R., and R. J. Byrd (2001), 'Knowledge portals and the emerging digital knowledge workplace', IBM Systems Journal Vol. 40, No. 4, 2001 - Knowledge Management: 925-955.

Mathiassen, L. and C. Sørensen (2002), A Task-Based Theory of Information Services. Proceedings of the Twenty-fifth Information Systems Research Seminar in Scandinavia, 2002.

Newell, S, H. Scarborough, J. Swan & D. Hislop, 'Intranets and knowledge management: complex processes and ironic outcomes', In: R. Sprague (ed.), Proceedings of the 32nd Annual Hawaii nternational Conference on System Sciences, Maui, Hawaii, 5-8 January 1999. IEEE, Los Alamitos, 1999.

Nielsen, J.: Intranet Portals: A Tool Metaphor for Corporate Information, Alertbox, March 31, 2003

Schöön, D.A, The reflective Practitioner, Basic Books, 1983.

Shilakes, C.C., and Tylman, J. (1998), Enterprise information portals: In-depth report, November 16, Merril Lynch, New York

Stenmark, D. (2002). "Information vs. Knowledge: The Role of intranets in Knowledge Management". In Proceedings of HICSS-35, Hawaii, January 7-10, 2002

Wenger, E. (1998), 'Communities of Practice. Learning, Meaning, and Identity', Cambridge: Cambridge University Press